Table 12. STM32G431RB 引脚定义(1)

LQFP64引脚号	引脚名(复位后功能)	引脚类型	I/O结构	注	可选功能	附加功能
1	VBAT	S	-	-	-	-
2	PC13	I/O	FT	(2) (3)	TIM1_BKIN,TIM1_CH1N,TIM8_CH4N,EVENTOUT	WKUP2,RTC_TAMP1,RTC_TS,RTC_OUT1
3	PC14-OSC32_IN	I/O	FT	(2) (3)	EVENTOUT	OSC32_IN
4	PC15-OSC32_OUT	I/O	FT	(2) (3)	EVENTOUT	OSC32_OUT
5	PF0-OSC_IN	I/O	FT_fa	-	I2C2_SDA,SPI2_NSS/I2S2_WS, TIM1_CH3N,EVENTOUT	ADC1_IN10,OSC_IN
6	PF1-OSC_OUT	I/O	FT_a	-	SPI2_SCK/I2S2_CK,EVENTOUT	ADC2_IN10,COMP3_INM,OSC_OUT
7	PG10-NRST	I/O	NRST (4)	-	MCO,EVENTOUT	NRST
8	PC0	I/O	FT_a	-	LPTIM1_IN1,TIM1_CH1,LPUART1_RX,EVENTOUT	ADC12_IN6,COMP3_INM
9	PC1	I/O	TT_a	-	LPTIM1_OUT,TIM1_CH2,LPUART1_TX,SAI1_SD_A, EVENTOUT	ADC12_IN7,COMP3_INP
10	PC2	I/O	FT_a	-	LPTIM1_IN2,TIM1_CH3,COMP3_OUT,EVENTOUT	ADC12_IN8
11	PC3	I/O	FT_a	-	LPTIM1_ETR,TIM1_CH4,SAI1_D1,TIM1_BKIN2,SAI1_SD_A,EVENTOUT	ADC12_IN9
12	PA0	I/O	TT_a	-	TIM2_CH1,USART2_CTS,COMP1_OUT,TIM8_BKIN, TIM8_ETR,TIM2_ETR, EVENTOUT	ADC12_IN1,COMP1_INM,COMP3_INP,RTC_TAMP2,WKUP1
13	PA1	I/O	TT_a	-	RTC_REFIN,TIM2_CH2,USART2_RTS_DE,TIM15_CH1N,EVENTOUT	ADC12_IN2,COMP1_INP,OPAMP1_VINP,OPAMP3_VINP
14	PA2	I/O	TT_a	-	TIM2_CH3,USART2_TX,COMP2_OUT,TIM15_CH1,LPUART1_TX, UCPD1_FRSTX,EVENTOUT	ADC1_IN3,COMP2_INM,OPAMP1_VOUT,WKUP4/LSCO
15	VSS	S	-	-	-	-
16	VDD	S	-	-	-	-
17	PA3	I/O	TT_a	-	TIM2_CH4, SAI1_CK1, USART2_RX,TIM15_CH2, LPUART1_RX, SAI1_MCLK_A, EVENTOUT	ADC1_IN4,COMP2_INP,OPAMP1_VINM/OPAMP1_VINP
18	PA4	I/O	TT_a	-	TIM3_CH2, SPI1_NSS, SPI3_NSS/I2S3_WS, USART2_CK, SAI1_FS_B,EVENTOUT	ADC2_IN17,DAC1_OUT1,COMP1_INM
19	PA5	I/O	TT_a	-	TIM2_CH1, TIM2_ETR,SPI1_SCK,UCPD1_FRSTX,EVENTOUT	ADC2_IN13,DAC1_OUT2,COMP2_INM,OPAMP2_VINM
20	PA6	I/O	TT_a	-	TIM16_CH1,TIM3_CH1, TIM8_BKIN,SPI1_MISO,TIM1_BKIN, COMP1_OUT,LPUART1_CTS,EVENTOUT	ADC2_IN3,OPAMP2_VOUT
21	PA7	I/O	TT_a	-	TIM17_CH1,TIM3_CH2,TIM8_CH1N,SPI1_MOSI,TIM1_CH1N,COMP2_OUT,UCPD1_FRSTX,EVENTOUT	ADC2_IN4,COMP2_INP,OPAMP1_VINP,OPAMP2_VINP
22	PC4	I/O	FT_fa	-	TIM1_ETR, I2C2_SCL, USART1_TX,EVENTOUT	ADC2_IN5
23	PC5	I/O	TT_a	-	TIM15_BKIN, SAI1_D3, TIM1_CH4N,USART1_RX, EVENTOUT	ADC2_IN11,OPAMP1_VINM,OPAMP2_VINM,WKUP5
24	PB0	I/O	TT_a	-	TIM3_CH3,TIM8_CH2N,TIM1_CH2N,UCPD1_FRSTX,EVENTOUT	ADC1_IN15,COMP4_INP,OPAMP2_VINP,OPAMP3_VINP
25	PB1	I/O	TT_a	-	TIM3_CH4, TIM8_CH3N, TIM1_CH3N,COMP4_OUT,LPUART1_RTS_DE,EVENTOUT	ADC1_IN12,COMP1_INP,OPAMP3_VOUT
26	PB2	I/O	TT_a	-	RTC_OUT2,LPTIM1_OUT,I2C3_SMBA,EVENTOUT	ADC2_IN12,COMP4_INM,OPAMP3_VINM
27	VSSA	S	-	-	-	-

28	VREF+	S	-	-	-	VREFBUF_OUT
29	VDDA	S	-	-	-	-
30	PB10	I/O	TT_a	-	TIM2_CH3,USART3_TX,LPUART1_RX,TIM1_BKIN,SAI1_SCK_A, EVENTOUT	OPAMP3_VINM
31	VSS	S	-	-	-	-
32	VDD	S	-	-	-	-
33	PB11	I/O	FT_a	-	TIM2_CH4,USART3_RX,LPUART1_TX,EVENTOUT	ADC12_IN14
34	PB12	I/O	FT_a	-	I2C2_SMBA,SPI2_NSS/I2S2_WS, TIM1_BKIN,USART3_CK,LPUART1_RTS_DE,EVENTOUT	ADC1_IN11
35	PB13	I/O	TT_a	-	SPI2_SCK/I2S2_CK,TIM1_CH1N,USART3_CTS,LPUART1_CTS,EVENTOUT	OPAMP3_VINP
36	PB14	I/O	TT_a	-	TIM15_CH1,SPI2_MISO,TIM1_CH2N,USART3_RTS_DE,COMP4_OUT,EVENTOUT	ADC1_IN5,OPAMP2_VINP
37	PB15	I/O	FT_a	-	RTC_REFIN,TIM15_CH2,TIM15_CH1N,COMP3_OUT,TIM1_CH3N, SPI2_MOSI/I2S2_SD,EVENTOUT	ADC2_IN15
38	PC6	I/O	FT	-	TIM3_CH1, TIM8_CH1,I2S2_MCK, EVENTOUT	-
39	PC7	I/O	FT	-	TIM3_CH2, TIM8_CH2, I2S3_MCK, EVENTOUT	-
40	PC8	I/O	FT_f	-	TIM3_CH3, TIM8_CH3, I2C3_SCL, EVENTOUT	-
41	PC9	I/O	FT_f	-	TIM3_CH4, TIM8_CH4, I2SCKIN, TIM8_BKIN2, I2C3_SDA, EVENTOUT	-
42	PA8	I/O	FT_f	-	MCO, I2C3_SCL,I2C2_SDA, I2S2_MCK,TIM1_CH1,USART1_CK, TIM4_ETR, SAI1_CK2,SAI1_SCK_A,EVENTOUT	-
43	PA9	I/O	FT_fd	-	I2C3_SMBA, I2C2_SCL,I2S3_MCK, TIM1_CH2,USART1_TX, TIM15_BKIN,TIM2_CH3, SAI1_FS_A,EVENTOUT	UCPD1_DBCC1
44	PA10	I/O	FT_da	-	TIM17_BKIN,USB_CRS_SYNC,I2C2_SMBA,SPI2_MISO, TIM1_CH3,USART1_RX,TIM2_CH4, TIM8_BKIN,SAI1_D1, SAI1_SD_A,EVENTOUT	UCPD1_DBCC2
45	PA11	I/O	FT_u	-	SPI2_MOSI/I2S2_SD,TIM1_CH1N,USART1_CTS,COMP1_OUT,FDCAN1_RX,TIM4_CH1,TIM1_CH4,TIM1_BKIN2,EVENTOUT	USB_DM
46	PA12	I/O	FT_u	-	TIM16_CH1,I2SCKIN,TIM1_CH2N,USART1_RTS_DE,COMP2_OUT,FDCAN1_TX,TIM4_CH2,TIM1_ETR,EVENTOUT	USB_DP
47	VSS	S	-	-	-	-
48	VDD	S	-	-	-	-
49	PA13	I/O	FT_f	(5)	SWDIO-JTMS, TIM16_CH1N, I2C1_SCL, R_OUT,USART3_CTS,TIM4_CH3,SAI1_SD_B,EVENTOUT	-
50	PA14	I/O	FT_f	(5)	SWCLK-JTCK,LPTIM1_OUT,I2C1_SDA,TIM8_CH2,TIM1_BKIN, USART2_TX,SAI1_FS_B,EVENTOUT	-
51	PA15	I/O	FT_f	(5)	JTDI,TIM2_CH1, TIM8_CH1, I2C1_SCL, SPI1_NSS,SPI3_NSS/ I2S3_WS, USART2_RX,UART4_RTS_DE,TIM1_BKIN, TIM2_ETR, EVENTOUT	-
52	PC10	I/O	FT	-	TIM8_CH1N,UART4_TX,SPI3_SCK/I2S3_CK,USART3_TX,EVENTOUT	-
53	PC11	I/O	FT_f	-	TIM8_CH2N,UART4_RX,SPI3_MISO,USART3_RX, I2C3_SDA, EVENTOUT	-
54	PC12	I/O	FT	-	TIM8_CH3N,SPI3_MOSI/I2S3_SD,USART3_CK,UCPD1_FRSTX,EVENTOUT	-

55	PD2	I/O	FT	-	TIM3_ETR, TIM8_BKIN,EVENTOUT	-
56	PB3	I/O	FT	(5)	JTDO-TRACESWO,TIM2_CH2, TIM4_ETR,USB_CRS_SYNC, TIM8_CH1N, SPI1_SCK, SPI3_SCK/I2S3_CK, USART2_TX,TIM3_ETR, SAI1_SCK_B,EVENTOUT	-
57	PB4	I/O	FT_c	(6)	JTRST,TIM16_CH1,TIM3_CH1,TIM8_CH2N,SPI1_MISO, SPI3_MISO,USART2_RX,TIM17_BKIN,SAI1_MCLK_B, EVENTOUT	UCPD1_CC2
58	PB5	I/O	FT_f	-	TIM16_BKIN, TIM3_CH2, TIM8_CH3N, I2C1_SMBA, SPI1_MOSI, SPI3_MOSI/I2S3_SD,USART2_CK,I2C3_SDA, TIM17_CH1, LPTIM1_IN1,SAI1_SD_B,EVENTOUT	-
59	PB6	I/O	FT_c	(6)	TIM16_CH1N,TIM4_CH1, TIM8_CH1,TIM8_ETR, USART1_TX,COMP4_OUT,TIM8_BKIN2,LPTIM1_ETR, SAI1_FS_B,EVENTOUT	UCPD1_CC1
60	PB7	I/O	FT_f	-	TIM17_CH1N,TIM4_CH2, I2C1_SDA,TIM8_BKIN, USART1_RX,COMP3_OUT,TIM3_CH4,LPTIM1_IN2, UART4_CTS,EVENTOUT	PVD_IN
61	PB8-BOOT0	I/O	FT_f	(7)	TIM16_CH1,TIM4_CH3, SAI1_CK1, I2C1_SCL, USART3_RX,COMP1_OUT,FDCAN1_RX,TIM8_CH2, TIM1_BKIN,SAI1_MCLK_A,EVENTOUT	-
62	PB9	I/O	FT_f	-	TIM17_CH1,TIM4_CH4,SAI1_D2, I2C1_SDA,IR_OUT, USART3_TX,COMP2_OUT,FDCAN1_TX,TIM8_CH3,TIM1_CH3N, SAI1_FS_A,EVENTOUT	-
63	VSS	S	-	-	-	-
64	VDD	S	-	-	-	-

1. Function availability depends on the chosen device.
2. PC13, PC14 and PC15 are supplied through the power switch. Since the switch only sinks a limited amount of current (3 mA), the use of GPIOs PC13 to PC15 in output mode is limited:
   - The speed should not exceed 2 MHz with a maximum load of 30 pF
   - These GPIOs must not be used as current sources (for instance to drive an LED).
3. After a Backup domain power-up, PC13, PC14 and PC15 operate as GPIOs. Their function then depends on the content of the RTC registers which are not reset by the system reset. For details on how to manage these GPIOs, refer to the Backup domain and RTC register descriptions in the reference manual RM0440 "STM32G4 Series advanced Arm®-based 32-bit MCUs".
4. PG10-NRST pin is FT tolerant if it is configured as PG10 GPIO by option bytes except for the startup time until option bytes are loaded.
5. After reset, these pins are configured as JTAG/SW debug alternate functions, and the internal pull-up on PA15, PA13, PB4 pins and the internal pull-down on PA14 pin are activated.
6. After reset, a pull-down resistor (Rd = 5.1kΩ from UCPD peripheral) can be activated on PB6, PB4 (UCPD1_CC1, UCPD1_CC2). The pull-down on PB6 (UCPD1_CC1) is activated by high level on PA9 (UCPD1_DBCC1). The pull-down on PB4 (UCPD1_CC2) is activated by high level on PA10 (UCPD1_DBCC2). This pull-down control (dead battery support on UCPD peripheral) can be disabled by setting bit UCPD1_DBDIS=1 in the PWR_CR3 register. PB4, PB6 have UCPD_CC functionality which implements an internal pull-down resistor (5.1kΩ) which is controlled by the voltage on the UCPD_DBCC pin (PA10, PA9). A high level on the UCPD_DBCC pin activates the pull-down on the UCPD_CC pin. The pull-down effect on the CC lines can be removed by using the bit UCPD1_DBDIS =1 (USB Type-C and power delivery dead battery disable) in the PWR_CR3 register.
7. It is recommended to set PB8 in another mode than analog mode after startup to limit consumption if the pin is left unconnected.